

Solutions HW #09

1. From HW #8, Prob. 6,

$$f^* = \sum_{i=1}^n \beta_i y_i$$

where

$$\begin{bmatrix} \langle y_1, y_1 \rangle & \dots & \langle y_1, y_p \rangle \\ \vdots & & \vdots \\ \langle y_p, y_1 \rangle & \dots & \langle y_p, y_p \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

(a) We have only 1 constraint, and

thus $f^* = \beta \cdot \underline{t}$ where

$$\langle t, t \rangle \beta = 2$$

$$\langle t, t \rangle = \int_0^2 \tau^2 d\tau = \left. \frac{\tau^3}{3} \right|_0^2 = \frac{8}{3}$$

$$\therefore \frac{8}{3} \cdot \beta = 2 \Rightarrow \beta = \frac{3}{4}$$

$$\therefore \boxed{f^* = \frac{3}{4} t}$$

(b) We have 2 constraints, and 2

thus

$$\begin{bmatrix} \langle t, t \rangle & \langle t, \sin(\pi t) \rangle \\ \langle t, \sin(\pi t) \rangle & \langle \sin(\pi t), \sin(\pi t) \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 2 \\ \pi \end{bmatrix}$$

$$\langle \sin(\pi t), \sin(\pi t) \rangle = \int_0^2 \sin^2(\pi \tau) d\tau = 1$$

$$\langle t, t \rangle = 8/3 \quad \text{from part (a)}$$

$$\langle t, \sin(\pi t) \rangle = \int_0^2 \tau \sin(\pi \tau) d\tau = -\frac{2}{\pi}$$

$$\therefore \begin{bmatrix} 8/3 & -2/\pi \\ -2/\pi & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 2 \\ \pi \end{bmatrix}$$

$$\beta_1 = \frac{3\pi^2}{2\pi^2 - 3}$$

$$\beta_2 = \frac{\pi(2\pi^2 + 3)}{2\pi^2 - 3}$$

$$f^* = \frac{3\pi^2}{2\pi^2 - 3} \cdot t + \frac{\pi(2\pi^2 + 3)}{2\pi^2 - 3} \sin(\pi t)$$

2. ^{Grizzle} ✓

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Will insert latex from the complete solution typed last week.

3. See ~~HW 2~~ Latex file!

4. See latex file.

~~5. From HW 2, Prob 7-(b), we know~~

~~that, for Q symmetric~~

~~$$\min_{x^T x = 1} x^T Q x = \lambda_{\min}(Q)$$~~

~~and~~

~~$$\arg \min_{x^T x = 1} x^T Q x = v_{\min}$$~~

5. In HW 2, Prob 7(L), we set $M = A^T A$, which is symmetric. We deduce that

$$\min_{x^T x = 1} x^T A^T A x = \lambda_{\min}(A^T A)$$

and

$$\arg \min_{x^T x = 1} x^T A^T A x = v_{\min}$$

where $v_{\min}$ is an e-vector of $A^T A$ corresponding to the e-value $\lambda_{\min}$; i.e., $A^T A v_{\min} = \lambda_{\min} v_{\min}$.

Moreover, the length of $v_{\min}$ must be chosen to be 1 to satisfy

$$v_{\min}^T v_{\min} = 1.$$

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From lecture, the columns of V are orthonormal e-vectors of $A^T A$, arranged so that the e-values of $A^T A$ are in decreasing order ∇ . Hence, $v_{\min}$ is the LAST column of V .

~~6. From the SVD handout, the smallest perturbation E such that the rank of $A+E$ is~~

~~2~~

6. Let $[U, S, V] = \text{svd}(A)$; that ⁶

is

$$A = U S V^T$$

with U and V orthogonal, S is diagonal. From page 13 of the SVD handout,

$$\hat{A} = U D V^T$$

where

$$D = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and

$$S = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}.$$

Doing the calculations in Matlab yields (see Prob 6.txt)

7.

(a) With $B = \text{double}(A)$ as in the problem statement, we do $[U, S, V] = \text{svd}(B)$. ~~Four images were generated using~~ The plot of the singular values, given by $\text{diag}(S)$, shows that the singular values decrease rapidly

Plot here.

SingularValues.png ← MATLAB folder

Four images were generated accounting for "100%", "90%",

§

"75%" and "50%" of the area under the curve. For example

```
>> IntegSingVal  
>> E = cumsum(diag(S));  
>> K = IntegSingVal(end);  
>> k2 = find(IntegSingVal > .9 * K);  
k2 = min(k2).
```

$$C_2 = U(:, 1:k_2) * S(1:k_2, :) * V(1:k_2, :)^{'}$$

Gives $k_2 = 99$ and the second image.

You do not need to do anything fancy! Just playing around is fine.

Include Four Zebra Images. png

(b). Anything is fine.